MOBILE UI DESIGN PROPOSAL

AEGIS BROWSER

Liquid Glass Dock

A zero-chrome concept for Aegis on Android: every page runs edge to edge at full screen, while the URL, new-tab and menu controls live inside one liquid-glass dock — and collapse to a five-pixel sliver on the left edge.

Concept v1.0 · Interaction & Visual Specification

Hardened by default. Quiet by design.

AEGIS-BROWSER.PAGES.DEV · NOVEMBER 2026

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01 · Design Concept — Zero Chrome

Aegis Browser is a privacy-hardened Chromium build for Android whose promise is already written into its identity: **hardened by default, quiet by design**. Yet on a typical phone, the browser itself refuses to be quiet. A conventional mobile layout spends a 56 dp toolbar on top, a 24 dp status strip and another 56 dp bar on the bottom — roughly 15% of a 412 × 924 dp viewport is furniture, not web. The user bought a tall, beautiful screen and receives a window with a mailing envelope taped to both ends.

This proposal inverts that arrangement. In the Aegis concept, **the page is the interface**: every website loads edge to edge, under the status bar and down to the navigation gesture area, exactly like a native app. There is no URL bar on top and no menu bar on the bottom. All browsing controls — the URL and trust indicator, a new-tab button and the three-dot menu — retreat into a single **liquid-glass dock** that floats above the page, and when they are not needed the dock collapses into a five-pixel sliver hugging the left edge of the screen. Chrome stops being a place and becomes a material: present when summoned, gone when reading.

3.6%	0.16%	92,957
of the viewport covered by the dock, expanded — versus ≈15% for a conventional top + bottom chrome layout	of the viewport covered by the collapsed edge sliver — effectively invisible while reading	ad and tracker hosts Aegis blocks at network level before first paint — trust rendered as a number, not a badge

The visual language is strictly monochrome — true blacks, paper whites and a calibrated grey ramp — because Aegis already ships a dark-first, monochrome, Montserrat-driven identity with Android 13 monochrome theming support. Liquid glass is the single material used across the dock, sheets and overlays: the live page is blurred and tinted *behind* the control, so the interface is literally made of the content it serves. In dark theme the glass is smoked ink; in light theme it is frosted paper. Both are rendered by the same recipe (Section 05), so the two themes are true parity, not an afterthought.

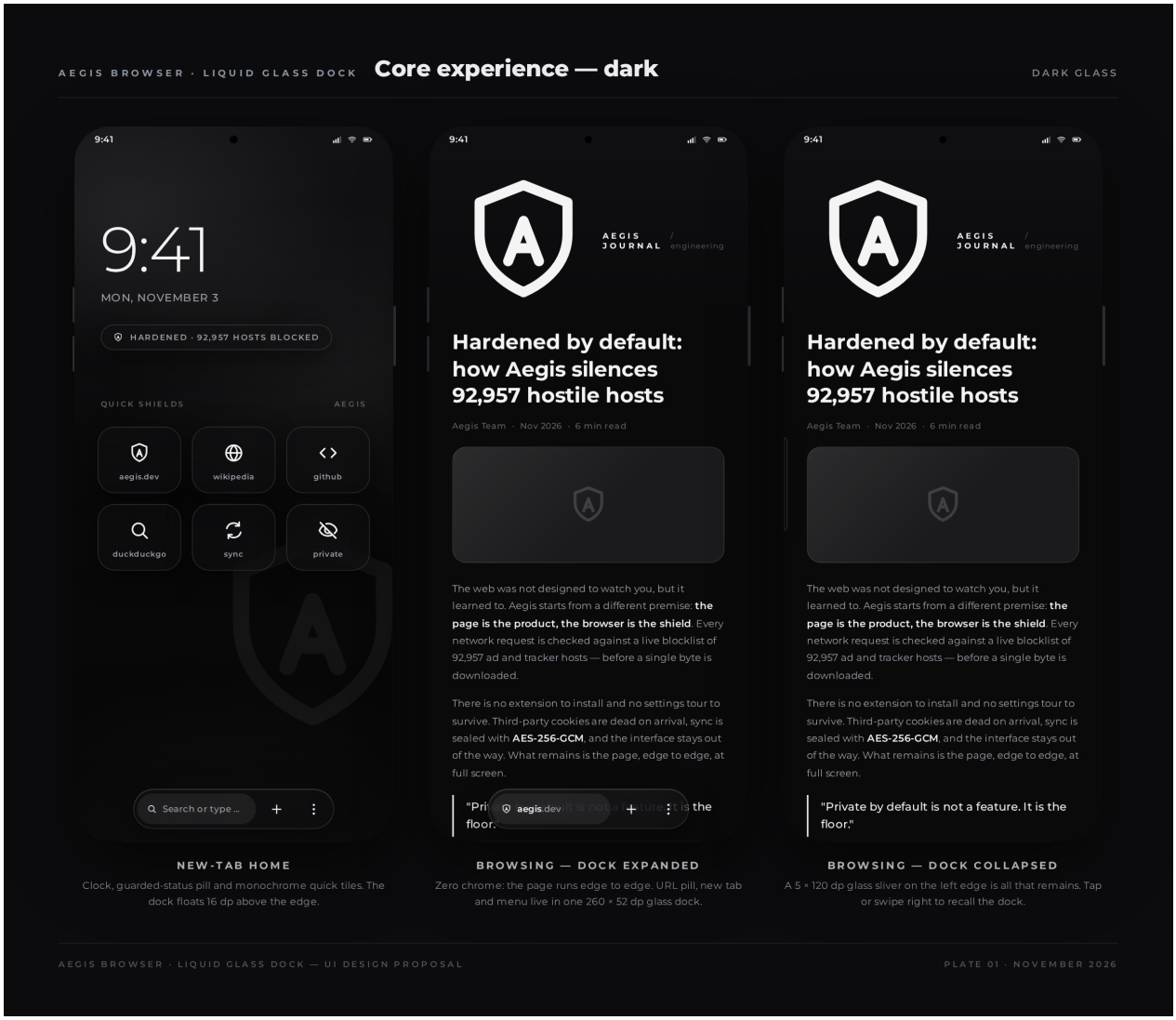


Plate 01 — Core experience, dark theme: new-tab home · browsing with the dock expanded · browsing collapsed to the left-edge sliver.

The result reads less like a browser and more like the site itself. On the new-tab home, the glass dock carries a search pill instead of a URL; while browsing, the same dock morphs to show the domain with the shield count. Everything else — tab switching, private mode, search engines — opens as glass sheets over the untouched page.

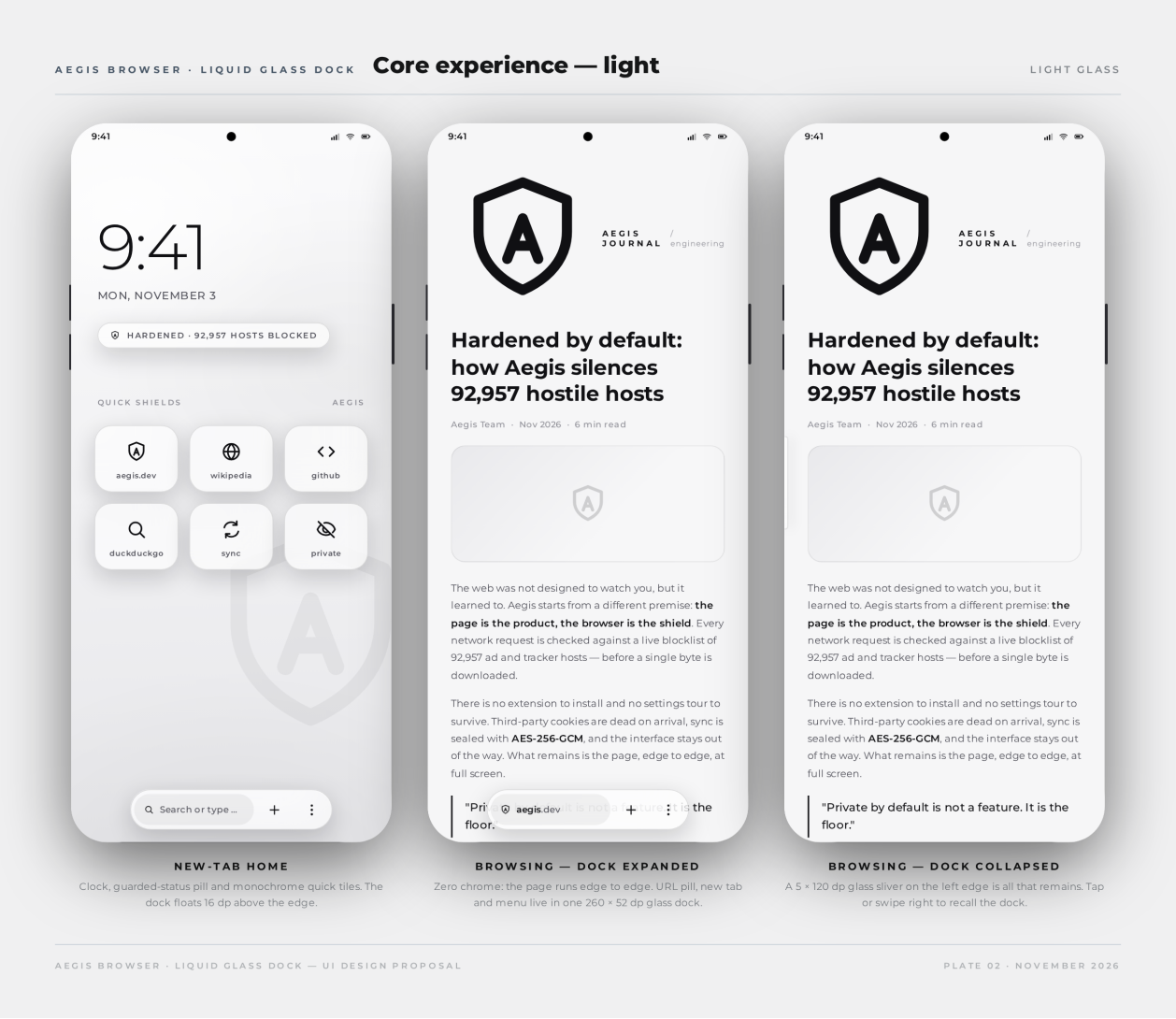


Plate 02 — Core experience, light theme: identical geometry, frosted-paper glass. Every state in this proposal is specified for both themes.

02 · The Liquid Glass Dock

The dock is the entire browser chrome compressed into one pill: **260 × 52 dp**, floating 16 dp above the bottom edge and horizontally centered. It carries exactly three controls, in reading order of frequency: the **URL and trust pill** (tap to search or edit the address), the **new-tab button**, and the **three-dot menu**. Nothing else earns a place on the first level. The pill is a true capsule — corner radius equal to half its height — so it reads as a droplet resting on the page rather than a bar bolted underneath it.

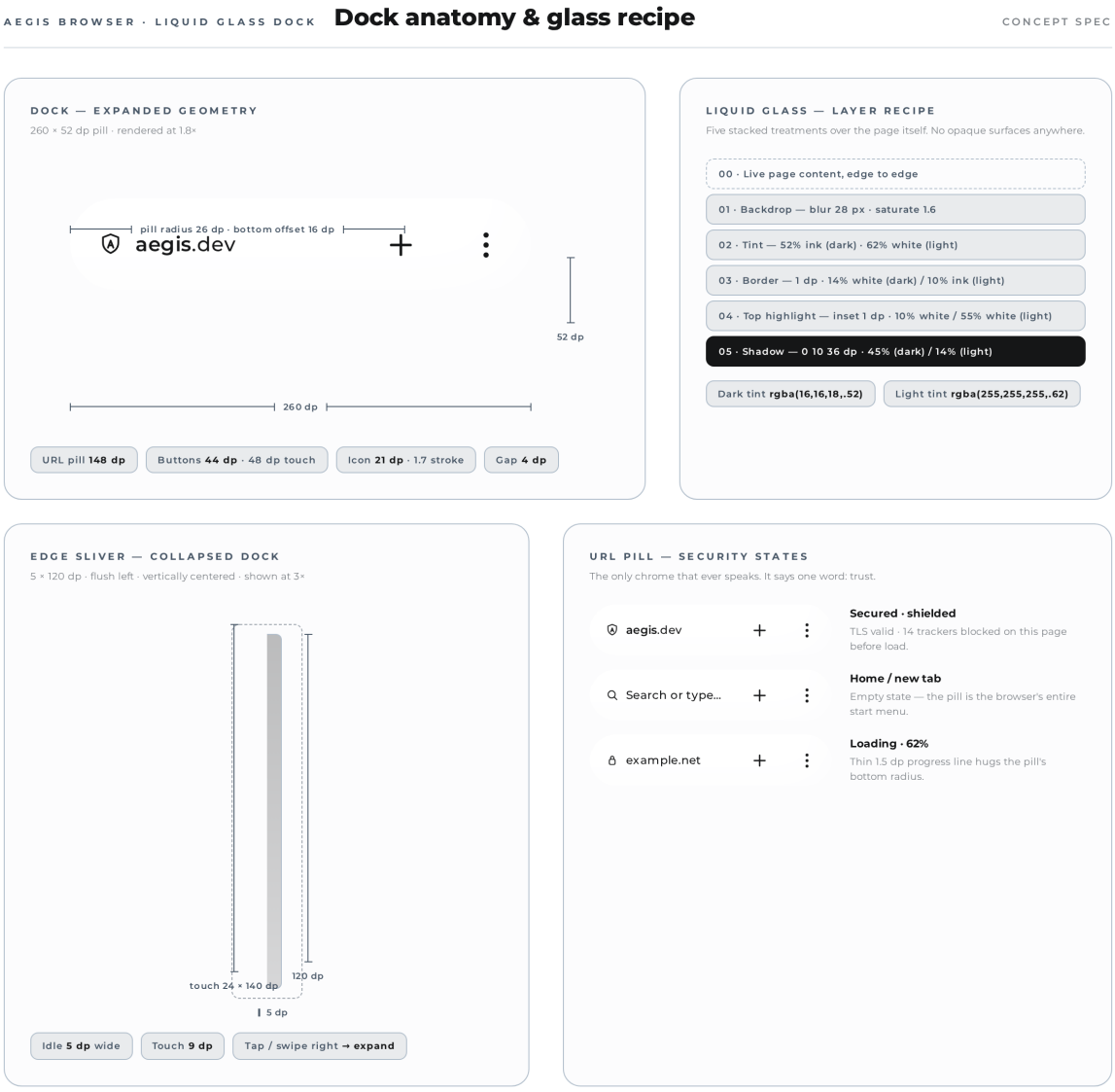


Plate 03 — Dock anatomy at 1.8× and the sliver at 3×: dimensions, the five-layer glass recipe, and the three URL-pill trust states.

Table 1 — Expanded dock geometry

Element	Specification	Notes
Container	260 × 52 dp, radius 26 dp	Pill; floats 16 dp above bottom, centered
URL / trust pill	148 × 40 dp, radius 20 dp	Shield glyph + domain, e.g. "aegis.dev"
New-tab button	44 × 44 dp circle	48 dp touch target via outset hit area
Menu button (3-dot)	44 × 44 dp circle	Opens the menu sheet (Section 04)
Iconography	21 dp glyph, 1.7 dp stroke	Monochrome outline set, round caps
Spacing	4 dp internal gaps, 4 dp side padding	Breathing room inside the capsule

All values in density-independent pixels at the 412 × 924 dp baseline.

The glass itself is a five-layer recipe applied over the live page: a 28 px backdrop blur with 1.6× saturation, a monochrome tint (52% ink in dark theme, 62% paper-white in light theme), a 1 dp hairline border, a 1 dp inner top highlight that catches light along the upper radius, and a soft drop shadow. No layer is opaque; the page always remains faintly visible through the control, which is what makes the dock feel liquid rather than painted. The full token values are specified in Section 05.

2.1 The edge sliver — collapsed state

When the user starts reading, the dock leaves. A flick down or left flings it off-screen and a **5 × 120 dp sliver** fades in, flush against the left edge and vertically centered. The sliver is the dock's ghost: the same glass recipe, one-pixel presence. Its touch target is deliberately generous — 24 × 140 dp — so a thumb can recall the full dock with a tap or a short swipe right, even though the visible target is nearly invisible. While a finger approaches (proximity or first touch), the sliver swells from 5 dp to 9 dp as a live affordance.

Table 2 — Edge sliver specification

Property	Value	Notes
Size (idle)	5 × 120 dp	Rounded 4 dp on the inner corners, flush left
Size (touch)	9 × 120 dp	Proximity swell, 120 ms ease-out
Position	left: 0, vertically centered	Anchored inside the gesture area, above nav inset
Touch target	24 × 140 dp	Meets accessibility minimum despite 5 dp visuals
Recall gestures	Tap; swipe right; long-press drag	Expand restores the dock with a 420 ms spring
Auto behavior	Appears 240 ms after dock dismissal	Staggered so the dock reads as flowing into the edge

The sliver inherits the dock's glass recipe unchanged — one material, two scales.

2.2 The trust line

Because the dock is the only chrome, the URL pill is the only place the browser is allowed to speak. It speaks about trust and nothing else: a shield glyph with the domain when the connection is secure, a loading state with a hairline progress curve hugging the pill's bottom radius, and an explicit "not encrypted" treatment for plain HTTP. On the new-tab home the same pill becomes "Search or type URL", so the dock doubles as the browser's entire start menu. Trackers blocked on the current page surface as a count inside the menu sheet header rather than as a permanent badge — reassurance on demand, not ambient anxiety.

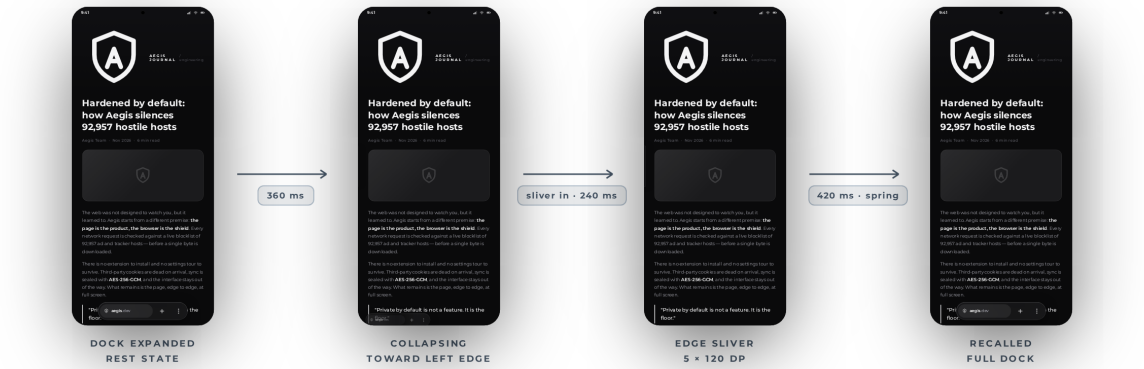
03 · Motion, Gestures and Auto-Hide

Motion in Aegis has one job: move chrome out of the way without making the user wait for it. Every transition completes in under half a second, every curve belongs to the Android "emphasized" family — fast departure, gentle landing — and nothing ever blocks the page. The dock's signature move is its collapse: the capsule shrinks, drifts left and thins into the sliver over 360 ms, so the state change reads as one continuous material flowing into the edge rather than one object leaving and another arriving.

AEGIS BROWSER · LIQUID GLASS DOCK

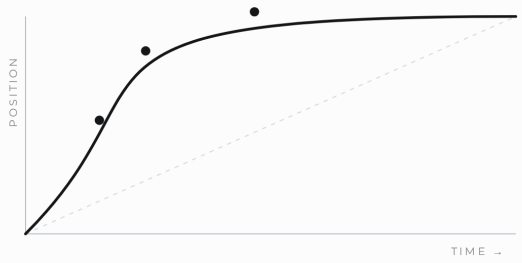
Motion & state transitions

CONCEPT SPEC



COLLAPSE CURVE — CUBIC-BEZIER(.32, .72, 0, 1)

Android "emphasized" family: fast departure, glass settles without bounce.



TIMING MAP

Every transition lands under half a second. Nothing blocks the page.

TRANSITION	DURATION	CURVE
Collapse to sliver	**360 ms**	cubic-bezier(.32, .72, 0, 1)
Expand from sliver	**420 ms**	cubic-bezier(.34, 1.28, .44, 1) — 4 dp overshoot
Overlay sheet rise	**380 ms**	cubic-bezier(.05, .7, .1, 1)
Scrim fade	**240 ms**	linear-out
Theme cross-fade	**250 ms**	ease
URL pill focus morph	**300 ms**	emphasized decelerate

AUTO-HIDE TRIGGERS

The dock manages itself; hands stay on the page.

- Scroll down · idle **1.2 s** → collapse
- Navigation complete → **peek 2.4 s**, then collapse
- Tap sliver / swipe right → **expand**
- Fling dock down or left → **collapse**
- Focus a text field → **stay hidden**
- Setting: **reading mode** default on

AEGIS BROWSER · LIQUID GLASS DOCK — UI DESIGN PROPOSAL PLATE 06 · NOVEMBER 2026

Plate 04 — Motion spec: the collapse flow at reading speed, the emphasized collapse curve, the timing map, and the auto-hide triggers.

Table 3 — Gesture map

Gesture	Context	Action
Tap URL pill	Dock expanded	Open search / address overlay
Tap + button	Dock expanded	New tab (immediate, no animation gate)
Long-press +	Dock expanded	Menu: reopen closed tab · new private tab
Tap 3-dot	Dock expanded	Open menu sheet
Fling dock down / left	Dock expanded	Collapse to edge sliver
Tap or swipe right on sliver	Dock collapsed	Expand the dock
Swipe up from bottom edge	Page, dock hidden	Open tab grid
Two-finger horizontal swipe	Page	Switch to previous / next tab
Long-press dock background	Dock expanded	Drag to reposition; release left = collapse

Gestures never conflict with page-level pans: dock gestures begin on dock chrome or system edges only.

3.1 Auto-hide behavior

The dock manages itself. Scrolling down and pausing for 1.2 seconds collapses it; the first upward scroll or a tap on the sliver restores it. After each navigation completes, the dock performs a 2.4-second peek so the trust line is always readable for a fresh page, then it retreats again. Focusing a text field keeps the dock hidden so the keyboard owns the bottom edge, and a "reading mode" setting (default on) makes the collapse behavior permanent for users who never want to see chrome mid-article. All timings and triggers are charted on Plate 04.

04 · Overlays — Tabs, Menu, Search

Three glass sheets do the rest of the work, and all three obey the same physics: they rise from the dock or the bottom edge, they blur the live page behind them, and they dismiss with a downward fling. Because the sheets are made of the same liquid glass as the dock, the interface never introduces a second material — panels feel like the dock pulled open rather than dialogs dropped on top. Scrim is a 2 px blur plus a 46% ink wash (28% in light theme): enough to focus, never enough to hide the page's structure.

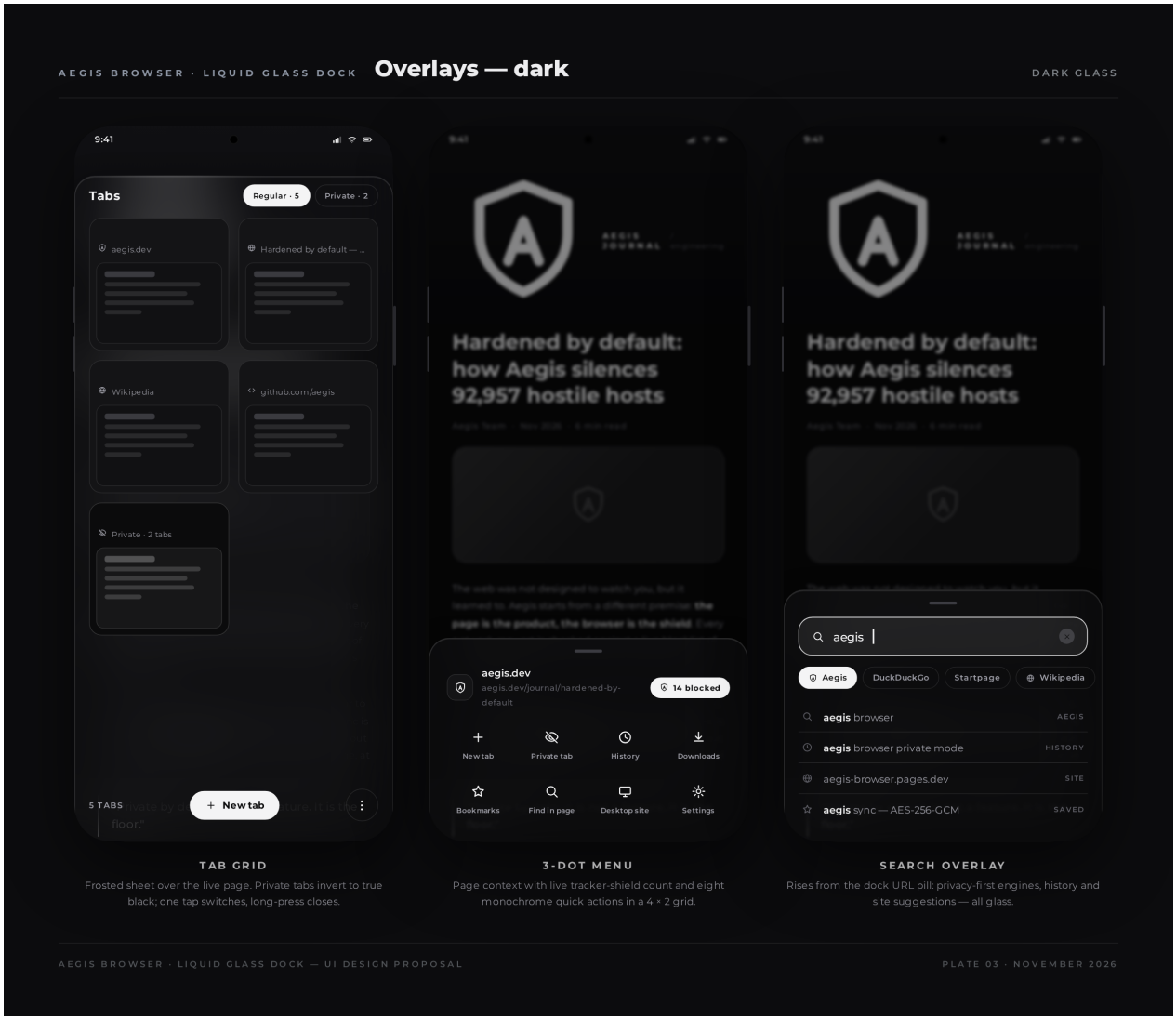


Plate 05 — Overlays, dark theme: tab grid with inverted private cards · 3-dot menu with live shield count · search overlay rising from the dock.

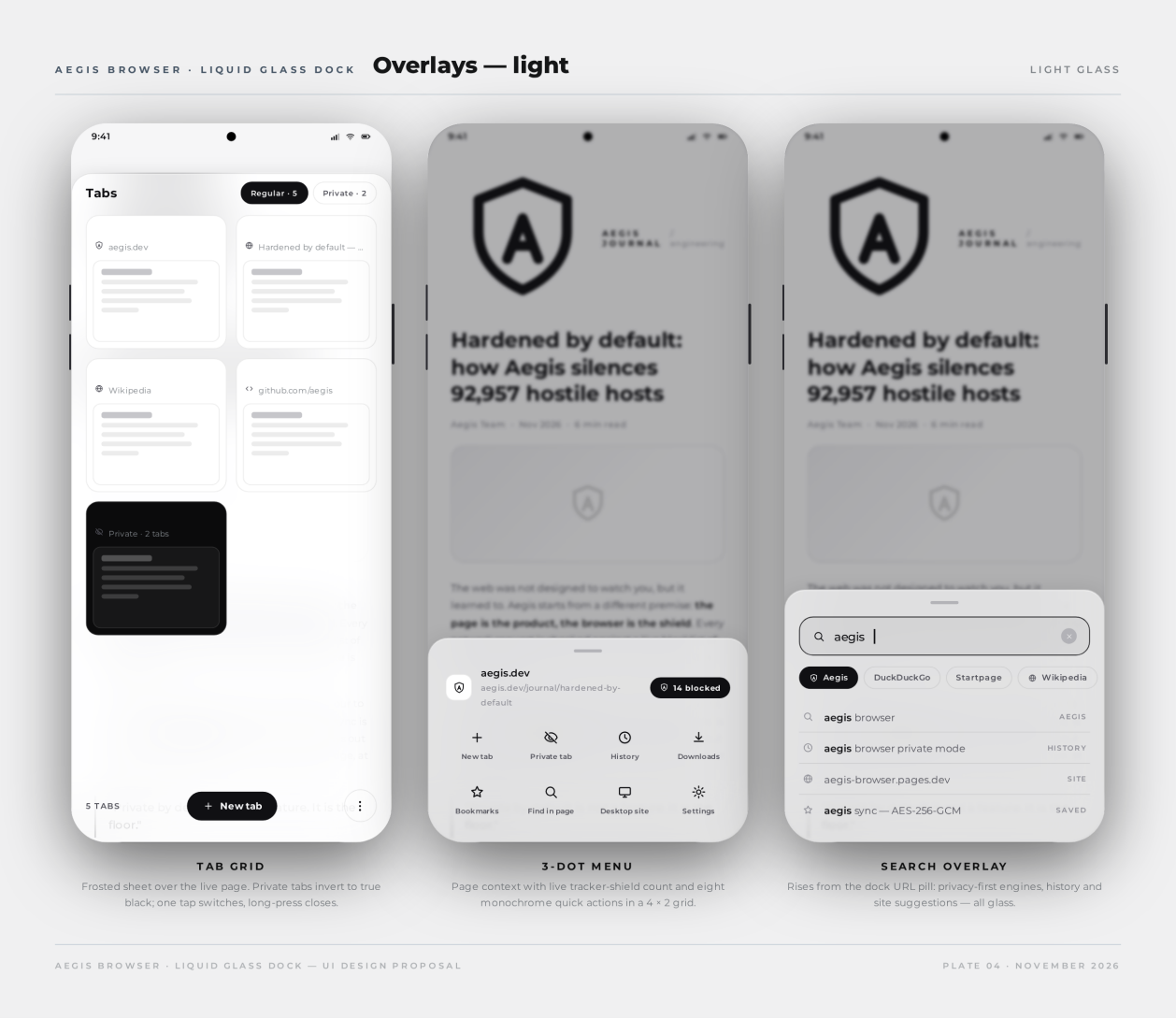


Plate 06 — Overlays, light theme: the same three sheets in frosted paper glass. Private tabs stay inverted in both themes.

Table 4 — Overlay specification

Overlay	Entry point	Contents	Dismiss
Tab grid	Swipe up from bottom edge; tab count in menu	2-column cards, favicon + title + live thumbnail; private cards inverted; "New tab" pill	Fling down; tap page area
3-dot menu	Dock menu button	Domain header with "N blocked" shield pill; 4 × 2 grid: new/private tab, history, downloads, bookmarks, find, desktop site, settings	Fling down; tap scrim
Search / URL	Tap URL pill	Focused input with caret; engine chips (Aegis, DuckDuckGo, Startpage, Wikipedia); history, site and saved suggestions	Fling down; system back

Sheets cap at 75% of viewport height; content longer than the cap scrolls inside the sheet.

05 · Design Tokens

Tokens are the contract between this proposal and the build. Every value below is theme-paired: dark first, light in lockstep, no orphan colors. The palette is intentionally achromatic — hierarchy is carried by lightness and blur, not hue — with pure white (dark theme) and pure ink (light theme) acting as the only "accent". This matches Aegis's existing monochrome identity and its Android 13 monochrome theming support, and it means the browser's chrome never competes with whatever colors the web page itself chooses to be.

Table 5 — Core palette

Token	Dark value	Light value	Usage
bg / page	#0A0A0B	#F6F6F7	Screen background behind content
surface	#141416	#FFFFFF	Cards, tab thumbnails
text / primary	#F4F4F5	#101013	Headings, active icons
text / secondary	#A1A1AA	#52525B	Body copy, labels
text / tertiary	#63636B	#8E8E96	Meta, captions, hints
line / hairline	white @ 10%	ink @ 8%	Dividers, card borders
accent (invert)	#FFFFFF	#0A0A0B	Primary buttons, active engine chip
scrim	ink @ 46%	ink @ 28%	Behind overlays

Greys form one ramp; both themes pass AA contrast for body text on their surfaces.

Table 6 — Liquid glass recipe

Layer	Dark theme	Light theme
Backdrop blur	28 px, saturate 1.6, brightness 1.06	28 px, saturate 1.4
Tint	rgba(16, 16, 18, 0.52)	rgba(255, 255, 255, 0.62)
Border	1 dp, white @ 14%	1 dp, ink @ 12%
Top highlight	inset 1 dp, white @ 10%	inset 1 dp, white @ 55%
Shadow	0 10 px 36 px, black @ 45%	0 10 px 30 px, ink @ 18%

Applied identically to dock, sliver, sheets and tiles — one material everywhere.

Table 7 — Typography (Montserrat)

Role	Spec	Usage
Display	82 dp / 300	Home clock
Page title (web)	27 dp / 700 · 1.28	Article H1 as rendered by reader styles
Overlay title	16 dp / 700	Sheet headers ("Tabs", domain)
Body / UI	13 dp / 400 · 1.7	Menus, suggestions, labels
Micro label	10 dp / 600 · +8% tracking, caps	Quick-shields header, engine chips
Pill text	11.5 dp / 500	URL and trust pill

Montserrat is the single UI family, matching the Aegis site; tabular figures for the clock.

Table 8 — Shape, spacing, elevation

Token	Value	Notes
Radius / dock	26 dp (full pill)	Half of 52 dp height
Radius / sheet	26 dp top corners	Bottom flush with screen edge
Radius / cards & tiles	16–22 dp	Tabs 16, home tiles 22, inputs 18
Spacing unit	4 dp base · 8/12/16/24 steps	Dock gaps 4, screen margins 30
Elevation 0	Glass only	Dock, sliver, tiles
Elevation 1	Glass + shadow 36 px blur	Expanded dock over scrolled content
Elevation 2	Glass + shadow + scrim	Menu and search sheets

Iconography: one outline set, 21 dp glyphs, 1.7 dp round-capped strokes, pure currentColor.

06 · Implementation Notes

The concept is deliberately buildable inside a Chromium codebase. The dock, sliver and sheets map to a single Android **View hierarchy overlay** (or Jetpack Compose layer hosted in the existing Activity) drawn above the WebContents frame — no changes to the rendering pipeline are required. The existing omnibox, tab-model and menu code paths are reused behind new presentation: the URL pill binds to the current Tab's security state, the tab grid binds to the TabModel filter (regular / private), and the menu sheet re-uses existing menu item ids. Edge-to-edge drawing uses WindowInsetsCompat with display-cutout passthrough, so the page truly reaches all four edges on punch-hole devices.

Listing 1 — Glass recipe as CSS custom properties (prototype reference for the Compose/View port)

```
/* Liquid glass — one recipe, both themes (tokens from Section 05) */

.glass {

  background: var(--glass-tint);           /* ink 52% · paper 62% */

  -webkit-backdrop-filter: blur(28px) saturate(1.6) brightness(1.06);

          backdrop-filter: blur(28px) saturate(1.6) brightness(1.06);

  border: 1px solid var(--glass-border);   /* white 14% · ink 12% */

  box-shadow: 0 10px 36px var(--glass-shadow),

              inset 0 1px 0 var(--glass-highlight);

  border-radius: 26dp;                     /* full pill at 52dp height */

}

/* Fallback: pre-render a blurred snapshot of the page (see Table 9) */
```

Table 9 — backdrop-filter fallback ladder

Condition	Strategy
GPU budget healthy (default)	Live backdrop-filter blur — true liquid glass
Battery saver / low-end device	Static blurred snapshot of the page, refreshed on scroll idle (200 ms throttle)
blur unsupported (old WebView)	Solid tint at 78% opacity, hairline border kept — dock remains fully legible
Accessibility "reduce transparency"	Tint 92%, blur off, borders strengthened

The dock must never render as a mystery rectangle: every tier keeps the pill silhouette and hairline.

Table 10 — Touch targets and accessibility

Item	Requirement
Dock buttons	44 dp visual, 48 dp touch target via outset hit area
Edge sliver	24 × 140 dp touch target despite 5 dp visuals; TalkBack announces "Show browser controls"
Sheet items	Minimum 48 dp row height; 4 × 2 menu grid tiles 88 dp wide
Contrast	Pill text 4.5:1 on glass in both themes (tint tuned to pass over any page)
Motion	Respects system "remove animations": transitions collapse to 80 ms fades
Focus order	Dock → sheet content → scrim dismiss; single back-stream integration

Targets follow Android accessibility guidance; glass never carries meaning without text or glyph.

Table 11 — Z-layer stack

z	Layer	Contents
0	WebContents	Live page, edge to edge, under status bar and cutout
1	Page UI	Find-in-page bar, HTML dialogs (page-owned)
2	Dock / sliver	Liquid glass chrome, elevation 0–1
3	Sheets + scrim	Tabs, menu, search; dim + blur page behind
4	System	Status bar icons, gesture navigation area, keyboard

Keyboard (layer 4) always wins the bottom edge; the dock yields and stays hidden while it is open.

6.1 Ship sequence at a glance

Stage	Scope	Done when
M1 · Dock	Expanded dock, bound URL pill, new tab, menu entry points	Dock renders over any page at 60 fps; omnibox round-trips
M2 · Collapse	Auto-hide triggers, collapse motion, edge sliver with recall gestures	Reading a long article shows only the sliver; recall is one tap
M3 · Sheets	Tab grid, menu sheet, search overlay on the shared glass sheet	All three open and dismiss under 400 ms with live page behind
M4 · Parity	Light theme, reduced-transparency and low-end fallback tiers	Token swap only; QA passes both themes and all fallback tiers

Each milestone ships on its own; nothing here requires compositor-level changes.

A pragmatic build order falls out of this stack: (1) the expanded dock with a bound URL pill, since it is the smallest slice that already feels like Aegis; (2) the collapse motion and edge sliver, which unlock the full-screen reading story; (3) the three glass sheets on the shared sheet component; (4) the light theme and reduced-transparency tiers, which are token swaps rather than new code. Each stage is independently shippable, and none of them requires touching the compositor — the entire proposal lives in the application layer, which is exactly where a hardened browser should keep its interface decisions.